

Design and Implementation of an IoT-Based Smart Waste Monitoring and Collection System Using ESP32

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Abstract— Modern cities have significant obstacles to effective waste collection and management due to rapid urbanization and rising waste creation. Conventional waste management systems have limited area coverage and inefficient operations since they rely on manual supervision and set timetables. This paper describes the design and implementation of an ESP32-based IoT-based smart garbage monitoring and collecting system to get around these restrictions. To provide real-time monitoring and communication, the suggested system integrates an ESP32 microcontroller with GPS and GSM modules and uses sensors at predetermined checkpoints. The movement of garbage collection vehicles and checkpoint coverage are recorded and visualized on a centralized software platform that receives location data and collection status continually. To facilitate effective decision-making, the system creates alarms, identifies missed milestones, and offers historical and real-time data via a software dashboard. When compared to traditional waste management techniques, experimental evaluation demonstrates that the suggested alternative increases collection efficiency, guarantees full area coverage, and minimizes manual involvement. The system is scalable, affordable, and ideal for waste management applications in smart cities.

Keywords— *Waste Collection Optimization, IoT, Smart Waste Management, ESP32, GPS, GSM, Real-Time Monitoring, and Smart Cities.*

I. INTRODUCTION

Solid waste output in cities has significantly increased due to rapid urbanization, rising income levels, and shifting consumption habits, making effective waste collection and management a top priority for municipal authorities. The majority of conventional waste management systems rely on manual monitoring and set collection schedules, which frequently result in ineffective routing, missed pickups, insufficient area coverage, and higher operating and environmental expenses. By connecting gadgets, automobiles, sensors, and infrastructure throughout urban settings, the Internet of Things (IoT) has become a crucial enabling technology in the context of smart city development, making cities safer, greener, and more effective. Real-time data collection and communication are made possible by IoT-based solutions, giving city authorities ongoing insight into the processes involved in waste generation and collection. Large amounts of spatial and temporal data are generated and processed by smart waste management systems using sensor-

equipped waste bins, GPS-enabled waste collection vehicles, and centralized software platforms. This data can then be analyzed to support dynamic decision-making, route optimization, checkpoint verification, and performance monitoring. However, the adoption of scalable and standards-based communication platforms and strong cooperation between many stakeholders, such as governments, system integrators, network operators, and technology vendors, are necessary for the successful implementation of such systems. The design and implementation of an Internet of Things (IoT)-based smart garbage monitoring and collecting system utilizing ESP32 is shown in this paper. It incorporates GPS and GSM modules for dependable data transmission and real-time vehicle tracking, as well as checkpoint-based monitoring to guarantee full area coverage. To display collection progress, identify missed checkpoints, and facilitate effective operational management, a centralized software dashboard is created. The suggested solution seeks to offer a realistic, scalable, and affordable method of smart trash management, supporting sustainable urban growth and raising the standard of living in smart cities.

The efficiency of smart waste management technologies in real-world deployments is still limited by a number of practical issues, despite their availability. The accountability and coverage of garbage collection operations are not sufficiently addressed by many of the current systems, which merely monitor the fill level of waste containers. Because of this, some locations or checkpoints might be overlooked during collection, which would result in overflowing bins, unsanitary conditions, and discontent from the general public. Furthermore, authorities find it challenging to confirm route completion and quickly address operational problems due to the absence of real-time monitoring of waste collection vehicles. The adoption of modern waste management technology is further hampered by high implementation costs, limited scalability, and reliance on constant internet access, particularly in emerging urban areas. With modularity and scalability in mind, the suggested IoT-based smart waste monitoring and collection system may be readily tailored to various urban settings and operational needs. Low power consumption, cost effectiveness, and flexibility in incorporating extra sensors and communication modules are guaranteed by using the ESP32 as the core controller. By providing a dependable, data-driven solution that promotes

sustainable urban growth and enhanced citizen quality of life, the system seeks to close the gap between theoretical smart city notions and real-world trash management applications.

II. LITERATURE SURVEY

Recent developments in data-driven technologies, artificial intelligence (AI), and the Internet of Things (IoT) have drastically changed conventional garbage management systems into intelligent and sustainable smart city solutions. Lakhout et al. [1] provided a thorough analysis of how AI and IoT are transforming urban solid waste management, emphasizing clever approaches to waste collection, sorting, and recycling. The study identified issues like scalability, cost, and data privacy while highlighting the usage of sensor-enabled waste bins, real-time data analytics, and intelligent decision-making to increase operational efficiency.

An Internet of Things (IoT)-based waste management system that integrates machine learning approaches to improve waste monitoring and optimization was proposed by Sumathi et al. [2] and implemented on a university campus. Their work showed the efficacy of integrating IoT infrastructure with cognitive analytics by demonstrating increased efficiency in trash level forecast and collection scheduling. In a similar vein, Anh Khoa et al. [14] investigated an Internet of Things (IoT)-based machine learning framework for waste management in educational institutions, concentrating on waste classification and monitoring to minimize human intervention and enhance campus sustainability.

With a focus on environmental sustainability, Ojadi et al. [3] expanded the use of AI-driven optimization to waste management and water use in smart cities. Their research demonstrated how resource-efficient urban management may be supported by AI-based optimization and predictive analytics. By suggesting clever operational management techniques, Kim et al. [4] tackled the clustered vehicle routing issue for waste collection and showed how optimal routing dramatically lowers fuel consumption, operating expenses, and collection time.

With a focus on modular system architecture, real-time dashboards, and maintainability, Alnanih et al. [7] demonstrated an IoT-driven smart trash management system with robust software engineering integration. Their research made clear how crucial software platforms are for providing municipal authorities with monitoring, transparency, and decision support. In their global bibliometric analysis of IoT-supported sustainable solid waste management, Zyoud and Zyoud et al. [8] identified research hotspots, technological trends, and gaps that point to an increasing demand for scalable and practically deployable systems.

The foundation for contemporary smart waste systems was laid by earlier pioneering work by Saha et al. [11], which showed the viability of IoT-based trash management employing sensors and wireless communication to monitor garbage levels. By combining deep learning and IoT to enable intelligent municipal trash management—with a specific emphasis on waste classification and automated decision-making—Wang et al. [9] substantially advanced this field.

Additionally, a number of research provide supporting technologies that are pertinent to smart waste systems. In their discussion of real-time optimal scheduling in IoT-based intelligent energy management systems, Rao et al. [5] offered insights into effective resource utilization and smart infrastructure-relevant decision algorithms. For scalable and low-latency applications, which are essential for real-time waste monitoring systems, Shai Kumar R et al. [6] examined cloud-edge-IoT collaboration. Sharu Shree et al. [16] examined lightweight encryption methods appropriate for energy-efficient IoT installations in order to meet the security and efficiency elements of IoT systems.

Saravanan et al. [12] presented a gesture-based mobile platform aimed at improving accessibility in educational environments through intuitive human–computer interaction. The system emphasized real-time sensor data processing, mobile application integration, and user-centered software design to enhance usability. Although the application domain differs from waste management, the study demonstrates effective integration of sensors with software platforms and real-time data handling. These concepts are relevant to IoT-based monitoring systems, where seamless interaction between hardware and software is essential. The software design approach discussed in [12] provides useful insights for the development of the centralized monitoring dashboard implemented in the proposed smart waste monitoring system by

The relevance of AI and advanced communication technologies in smart cities is also highlighted by recent studies. In their assessment of FANET–IoT–IoV interactions in 6G-enabled smart cities, Raza and Aadil [17] highlighted future connectivity prospects and problems. Further research on AI-driven systems [10], [13], [15], and IoT-based environmental monitoring [18] shows how intelligent data processing, secure communication, and sensor integration can be used in large-scale smart city settings.

III. RELATED WORK

The application of Internet of Things (IoT) technology to enhance the sustainability and effectiveness of urban solid waste management systems has been the subject of extensive research. Garbage cans with sensors to track fill levels and send status data to centralized monitoring platforms were the main focus of early IoT-based solutions. These technologies permitted rudimentary automation in waste collection procedures and effectively decreased manual inspection. However, their capabilities were mostly restricted to bin-level monitoring and did not offer enough support for vehicle tracking, real-time operational management, or confirmation of full coverage.

Artificial intelligence and machine learning techniques have recently been incorporated into intelligent waste management systems. AI-based methods have been used for waste classification, waste generation prediction, and collection schedule optimization. Through the analysis of both historical and current data, these systems improve decision-making accuracy and resource usage. Despite its efficacy, many AI-driven solutions are less suited for real-time

implementation in resource-constrained and cost-sensitive metropolitan areas because to their high computational resource requirements, vast datasets, and complicated infrastructure.

Route optimization and vehicle tracking for waste collection operations constitute another significant area of research. The goal of GPS-enabled waste collection vehicles in conjunction with clustering and routing algorithms is to reduce collection time, fuel consumption, and trip distance. Although these techniques increase operational efficiency, the majority of them focus mainly on route planning and optimization without offering real-time confirmation of whether all designated regions or checkpoints have been reached throughout the collecting process. Because of this, there is still little accountability and oversight of operations on the ground.

Additionally, a number of studies highlight how cloud-based and cloud-edge collaborative architectures facilitate centralized analytics, data storage, and monitoring. For municipal administrations, these technologies offer decision-support tools and real-time dashboards. Large-scale and practical implementation of such systems is still hampered by issues including latency, communication dependability, system integration complexity, and reliance on constant internet connectivity, especially in developing nations.

Research Gap: It is clear from the studied literature that current smart waste management solutions only handle certain areas, like data analytics, route optimization, and waste level monitoring. A unified system that guarantees continuous vehicle tracking, checkpoint-based verification of full area coverage, real-time monitoring of waste collection activities, and centralized software visualization in an economical and scalable way has received very little attention. Additionally, a lot of systems don't have workable ways to identify missed checkpoints and notify police right away. In order to close these gaps and ensure operational transparency, accountability, and increased efficiency for smart city waste management, this research proposes an ESP32-based IoT smart waste monitoring and collection system that combines GPS and GSM communication with checkpoint-based monitoring and a centralized software platform.

IV. PROPOSED SYSTEM

In order to guarantee effective, transparent, and responsible garbage collection in urban settings, the suggested system is an Internet of Things-based Smart garbage Monitoring and Collection System. The implemented hardware prototype illustrates how the system uses an ESP32 microcontroller as the core processor unit, integrated with a GPS module, a GSM communication module, and a battery-powered supply unit. The system's main goals are to send operational data to a centralized monitoring platform, confirm coverage of predetermined checkpoints, and track waste collection vehicles in real time.

$$D_r = N_p / T \quad (1)$$

Where:

- D_r = Data packets transmitted per second
- N_p = Number of packets sent
- T = Time duration

Waste collection routes are split up into several geographical checkpoints that cover every area that has been defined in the proposed architecture. The collection vehicle's latitude and longitude are continuously recorded by the GPS module. To determine if the vehicle has successfully visited each location, these coordinates are compared with checkpoint coordinates that have been stored. A checkpoint is designated as covered after it is reached within a predetermined tolerance radius. This guarantees that no region is overlooked when collecting rubbish.

$$\text{Coverage Efficiency (\%)} = N_t / N_c \times 100 \quad (2)$$

Where:

- N_c = Number of checkpoints covered
- N_t = Total number of checkpoints assigned

The ESP32 uses the GSM module to interact with the central server while processing sensor data and position information. Vehicle ID, current GPS coordinates, checkpoint status, timestamp, and battery status are among the data transmitted. The system can be continuously deployed in the field since it is powered by rechargeable lithium-ion batteries. Real-time vehicle movement and coverage status are visualized by the centralized software platform, which also notifies authorities of any missed checkpoints or unusual delays.

This integrated strategy is appropriate for smart city applications because it not only increases operational efficiency but also promotes accountability and transparency in municipal trash management.

$$D = (\sin(2\Delta\phi) + \cos(\phi_1) \cos(\phi_2) \sin(2\Delta\lambda)) \quad (3)$$

Where:

- ϕ_1, ϕ_2 = latitudes (in radians)
- λ_1, λ_2 = longitudes (in radians)
- $\Delta\phi = \phi_2 - \phi_1$
- $\Delta\lambda = \lambda_2 - \lambda_1$, if $d \leq R_{threshold}$ then the checkpoint is considered **covered**.

TABLE I. HARDWARE COMPONENTS

Components	Description	Purpose
ESP-32	32-bit microcontroller with Wi-Fi/Bluetooth	Data processing and control
GPS Module	NEO-6M or equivalent	Real-time location tracking
GSM	SIM800L	Data communication
Voltage Regulator	Buck converter	Stable power delivery
Li-on Batteries	18650 cells	Portable power supply

A. ESP32 Microcontroller

The suggested smart waste monitoring system's central control unit is the ESP32 microcontroller. All system functions, including sensor data collection, GPS data processing, checkpoint verification, and communication control, must be coordinated by it. The ESP32 was chosen because of its high processing power, low power consumption, and several serial communication interfaces (UART, SPI, and I2C) that enable smooth integration with GPS and GSM devices. It is ideal for dispersed IoT-based trash management systems because of its real-time processing capability, which permits quick decision-making and continuous vehicle location monitoring.



Fig 1. ESP – 32 Microcontroller

B. GPS Module

The waste collection vehicle's precise real-time positional data is obtained using the NEO-6M GPS module. computes geographic coordinates in terms of latitude and longitude after receiving signals from several satellites. Standard NMEA sentences are used in UART serial communication to send these coordinates to the ESP32. In order to determine if the car has arrived at predetermined checkpoint locations that correspond to waste collection places, the ESP32 evaluates the received GPS data. The NEO-6M module was chosen for continuous outdoor tracking applications because of its excellent sensitivity, quick time-to-TTFF, and low power consumption. In order to improve accountability and operational efficiency in municipal trash management, the GPS module is essential for exact route monitoring and position verification. This ensures that all garbage collection checkpoints are covered in a methodical manner.



Fig 2. NEO – 6M Module

C. SIM800L GSM Module

The suggested system has cellular communication capabilities thanks to the SIM800L GSM module. It makes it possible to provide real-time data to a centralized monitoring system or cloud server, including GPS coordinates, checkpoint status, timestamps, and alert notifications. Standard AT commands are used to establish communication between the GSM module and the ESP32 via UART. Wide network compatibility and dependable communication in both urban and semi-urban settings are guaranteed by the SIM800L's use of quad-band GSM frequencies. It is selected because of its small size, low power consumption, and reliable data transfer capabilities. The GSM module makes sure that system data can be accessed remotely without relying on local wireless infrastructure, allowing administrators or authorities to monitor waste collection operations in real time.



Fig 3. GPS Module – SIM 800L

D. Voltage Regulation Unit

All system components receive a steady and reliable power supply thanks to the voltage regulating unit. The voltage regulator guarantees appropriate voltage conversion and current stability because the ESP32 and GPS modules work

at various voltage levels and the SIM800L GSM module needs a greater peak current during transmission. By doing this, voltage swings that can result in a malfunctioning system or poor communication are avoided. During prolonged field operation, the regulating unit prolongs the life of electronic components and improves system reliability.



Fig 4. Buck Converter

E. Application Interface

The Smart Thuru application interface enables real-time monitoring and management of waste collection routes. It displays active routes, assigned collectors, and the progress of each route on a dynamic map. Collectors' locations are updated in real time, with starting points, pending collection points, and planned paths clearly marked. The interface allows administrators to track route completion, assign routes, and generate reports, providing an intuitive and efficient tool for operational management.

Key Points:

- Sidebar for navigation: Dashboard, Routes, Assign Routes, Track Routes, Collectors, Reports.
- Active routes panel shows collector name, route, and progress
- Dynamic route paths connect collection points with color-coded status.
- Real-time updates for route progress and collector status.
- User controls for refresh and profile/logout functions.

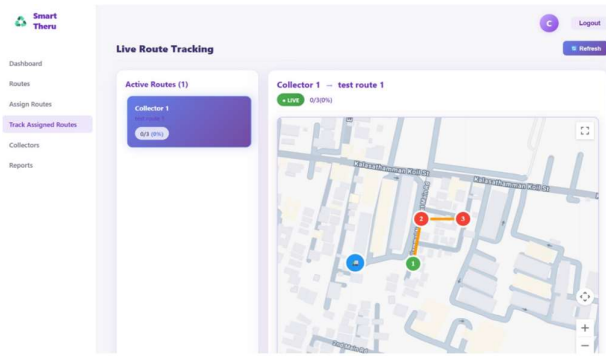


Fig 6. Live Dashboard of collection Management

V. SYSTEM ARCHITECTURE

The suggested smart trash monitoring system is structured into a layered approach to improve modularity, scalability, and operational clarity. Because each layer is in charge of a

distinct set of tasks, concerns are kept apart and system upkeep and future updates are made easier. The tiered architecture facilitates real-time monitoring and decision-making while enabling effective communication between software services and hardware components as layers as follows.

A. Device Layer

All of the physical hardware installed on the waste collecting vehicle makes up the device layer. This comprises voltage regulation circuitry, lithium-ion batteries, the ESP32 microprocessor, the NEO-6M GPS module, and the SIM800L GSM module. This layer is in charge of carrying out embedded operations, detecting real-world factors, and facilitating communication readiness. By supplying raw data and operational stability, it serves as the system's cornerstone.

B. Data Acquisition Layer

Real-time data collection from the device layer is the responsibility of the data acquisition layer. GPS coordinates, including latitude and longitude, are continuously obtained from the NEO-6M module via serial connection in the proposed system. This layer filters erroneous or incomplete readings before sending them for processing, ensuring accurate, timely, and consistent data collecting.

C. Processing and Decision Layer

All logical and computational functions of the ESP32 microcontroller are carried out by the processing and decision layer. It evaluates whether waste collection routes are correctly followed by analyzing incoming GPS data and comparing it with predetermined checkpoint locations. This layer performs threshold-based validation and distance calculations, allowing for real-time decision-making without reliance on outside sources.

D. Application and Monitoring Layer

The software interface that administrators and authorities employ is represented by the application and monitoring layer. Vehicle movement, checkpoint coverage, and system performance are all visualized when real-time data is received and stored in a database. This layer improves waste management efficiency and transparency by supporting operational oversight, reporting, and data-driven decision-making.

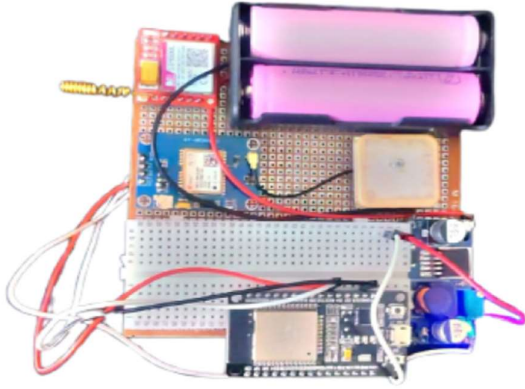


Fig 6. The Prototype of IoT-based smart garbage monitoring and collection system's hardware prototype

$$L_c = (\phi_c, \lambda_c) \quad (4)$$

Here:

- L_c represents the **central point** (maybe a centroid) in terms of latitude (ϕ_c) and longitude (λ_c).
- This could be the "reference" location against which other points are compared

$$L_i = (\phi_i, \lambda_i), i = 1, 2, \dots, N \quad (5)$$

Here:

- L_i represents the i -th **data point** (like a GPS location).
- ϕ_i is its latitude and λ_i is its longitude.
- N is the total number of points in your dataset.

$$D_i = (\sin^2(2\phi_i - \phi_c) + \cos(\phi_c)\cos(\phi_i)\sin^2(2\lambda_i - \lambda_c)) \quad (6)$$

The spatial deviation of point L_i from the center L_c is measured by D_i , which is defined by this equation. Let's examine each term individually:

- **First term:** $\sin^2(2\phi_i - \phi_c)$
 - Measures the squared sine of the **difference in latitude** (scaled by 2).
 - Captures how far north/south the point is relative to the center.
- **Second term:** $\cos(\phi_c)\cos(\phi_i)\sin^2(2\lambda_i - \lambda_c)$
 - Measures the squared sine of the **difference in longitude** (scaled by 2), weighted by the cosine of the latitudes.
 - Why weight with cosines? Because **longitudinal distances shrink as you move toward the poles.**
 - At the equator, the full longitude difference matters.

- Near the poles, small longitude differences correspond to very small actual distances.

- **Overall:** D_i is a kind of **distance metric on a sphere**, combining both latitude and longitude differences into a single scalar.

- Larger $D_i \rightarrow L_i$ is farther from the center L_c .
- Smaller $D_i \rightarrow L_i$ is closer to the center.

VI. FUTURE ENHANCEMENTS

Predictive analytics and machine learning algorithms will be included into the suggested IoT-based smart garbage monitoring and collection system in the future to forecast waste generation patterns and dynamically improve collection routes. To offer a more thorough understanding of trash types and amounts, the system can be enhanced by adding additional sensors, such as weight, chemical, or odor detectors. Citizens can be engaged, informed about bin statuses and collection schedules, and the amount of illegal dumping can be decreased by using real-time public announcements via web portals or mobile applications. Sustainability can be improved through energy-efficient operations that use low-power communication protocols and solar-powered bins. Additionally, the system can be transformed into a fully intelligent, predictive, and citizen-integrated smart waste management solution appropriate for future smart cities through integration with autonomous or semi-autonomous collection vehicles, cloud-based data analytics for long-term planning, and scalable deployment across multiple districts or cities.

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